Identify Testing Performance Disparity in the GED Marketplace

A data-driven analysis of demographic, geographic, and infrastructure drivers

The Problem We’re Solving

Context and key question

Situation

GED Testing Service helps adults earn a high school equivalency credential by passing four subject exams. Each year ~600K learners begin this journey, yet only ~35% progress to completing the exams.

Complication

Access to testing centers and educational support varies significantly across regions, potentially affecting candidates’ ability to prepare, test, and succeed. With limited education resources, GED must determine where disparities exist and where intervention would be most impactful.

Key Strategic Question

Where do testing performance disparities exist across candidate groups and geographic regions—and what factors drive these differences?

Sub-Questions We Explore

- Do GED results differ across different demographic groups (such as age, education level, language, or gender)?
- Do some states or regions perform better or worse on GED outcomes?
- How are testing resources distributed across regions?
- Are there areas where limited testing resources may be affecting candidate performance?
- Which demographic or regional factors appear to drive differences in GED outcomes?
- Can we identify demographic groups or regions that may need additional support to improve success rates?

Solution Map

1. **Exploratory Data Analysis (EDA)**
   Examine candidate demographics, performance metrics, and geographic characteristics.

2. **Resource Distribution Analysis**
   Analyze how testing resources are distributed across regions.

3. **Segmentation via Clustering**
   - Demographic clustering of candidates
   - Geographic clustering of states/regions

4. **Driver Identification**
   Analyze how infrastructure constraints and candidate characteristics relate to performance outcomes.

5. **Insights & Recommendations**
   Identify underserved segments and propose targeted resource allocation strategies.

01
Demographic Clustering Analysis

Demographic Factors Influence GED Test Outcomes

Performance Metrics

Average scores

The average test score across all test attempts

Passing rate

The percentage of test attempts that meet the passing standard

No-show rate

The percentage of registered test attempts where the examinee did not take the test

Key Demographic Factors

Education level

No Schooling
Elementary
Middle School
High School

Primary language

English
Spanish

Age

15-17, 18-24,
25-34, 35-44,
45-54, 55-64
65 +

Race Ethnicity

Races
Hispanic
Non-hispanic

Gender

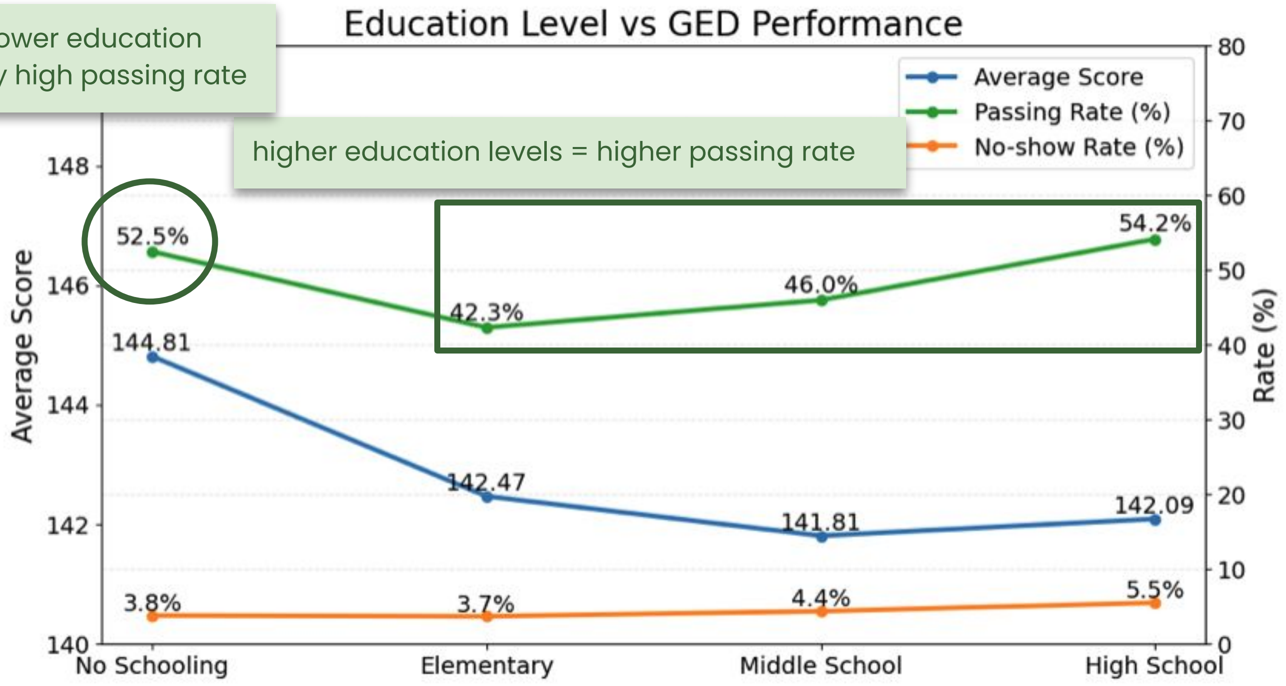
Male
Female
Nonbinary
Unknown

Demographic Factors Influence GED Test Outcomes

Preliminary Findings: Education Level V.S. GED Performance

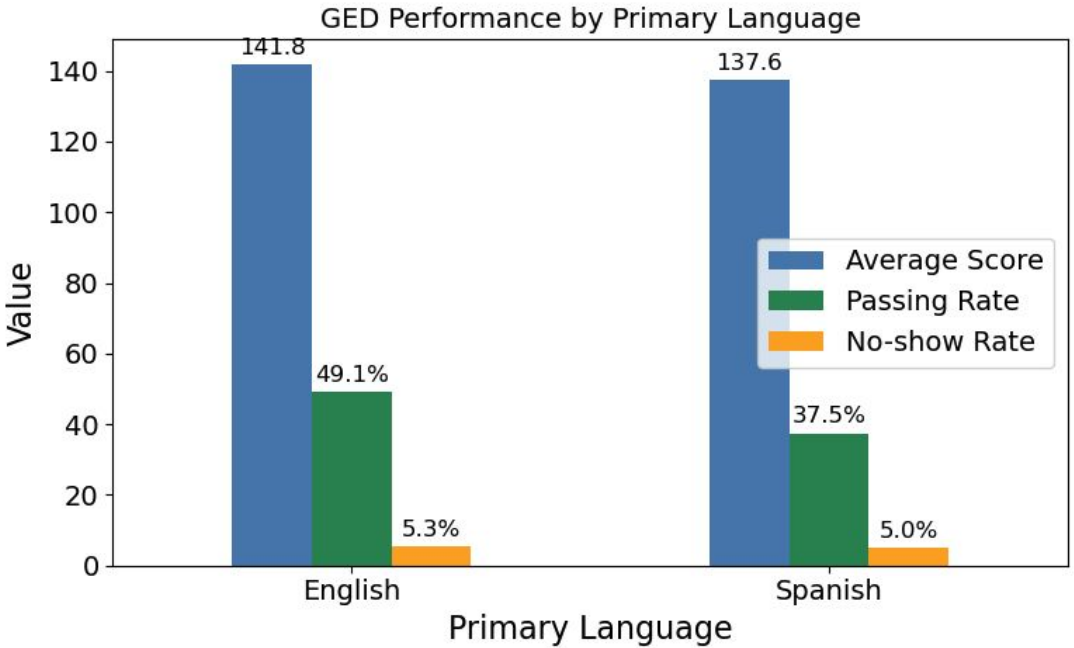
Unexpected pattern: lower education group shows relatively high passing rate

higher education levels = higher passing rate



Demographic Factors Influence GED Test Outcomes

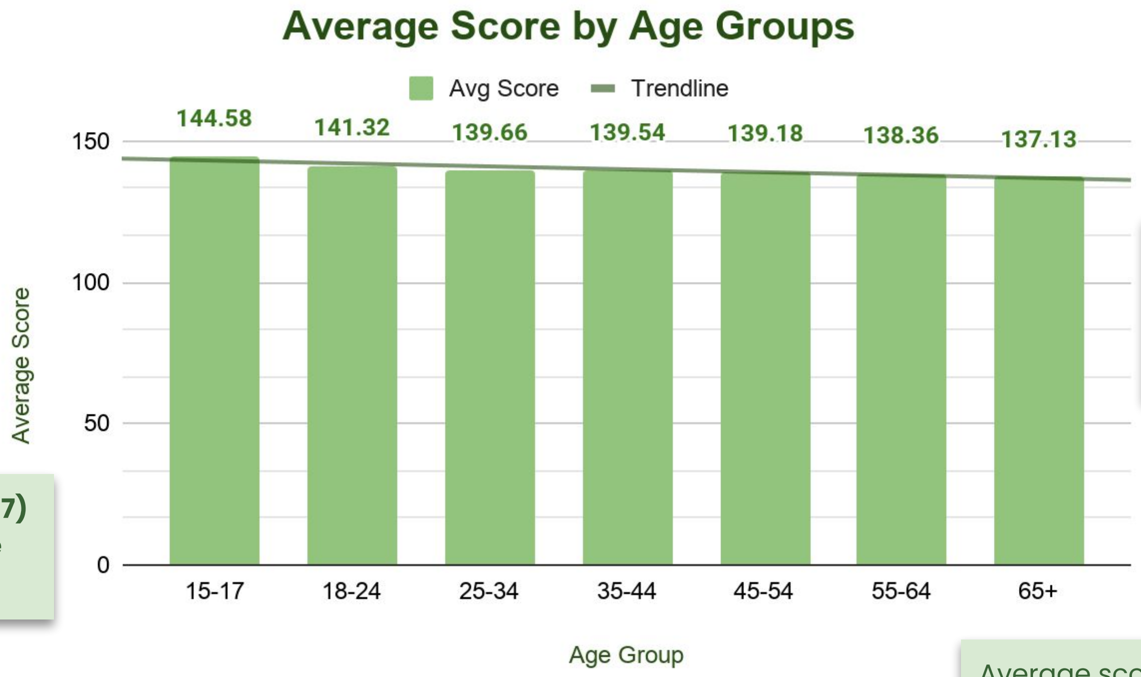
Preliminary Findings: Language code V.S. GED Performance



English test takers perform better than Spanish speakers, with similar no-show rates.

Demographic Factors Influence GED Test Outcomes

Preliminary Findings: Age V.S. GED Performance (Cont.)



Trendline shows a slight **negative** correlation between age and average GED score.

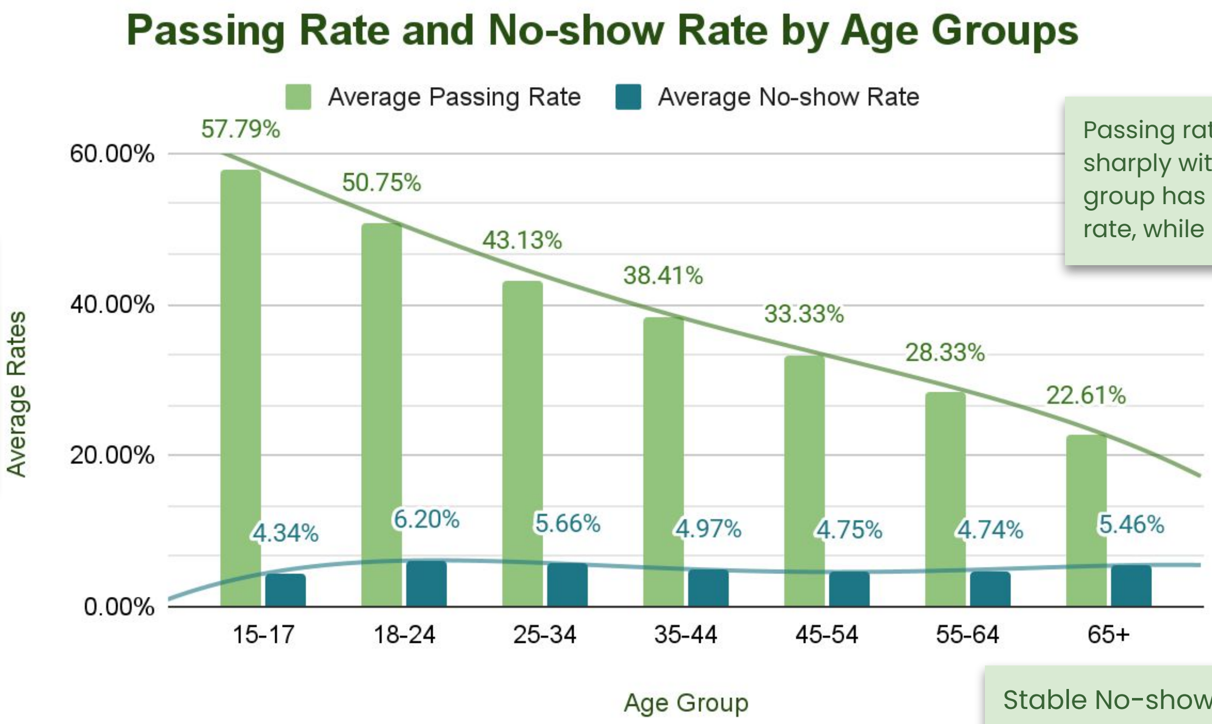
Younger test-takers **(15–17)** have the **highest** average scores **(~144.6)**.

Average scores **gradually decrease** with age, with 65+ scoring the lowest (~137.1).

Demographic Factors Influence GED Test Outcomes

Preliminary Findings: Age V.S. GED Performance (Cont.)

Age **negatively** affects GED performance: younger candidates perform better on average.



Passing rate **declines** sharply with age: 15–17 group has ~57.8% passing rate, while 65+ has ~22.6%.

Stable No-show rates (roughly 4–6%) across all ages , suggesting age does not impact no-show rates.

Demographic Factors Influence GED Test Outcomes

Four Distinct Demographic Segments with Varying GED Outcomes

Clustering Performed Using K-Modes

Variables used in clustering

Cluster	Language	Age Group	Race	Gender	Education	Ethnicity	Count	Count %	Avg Score	No-Show Rate %	Passing Rate %
Higher performing	English	18-24	White	Male	High School	Non-hispanic/ latino	776536	31.9	144.1	5.0	59.0
Moderate Performing	English	15-17	N/A	Male	High School	N/A	697046	28.6	142.2	5.1	52.8
Lower Performing	English	18-24	White	Female	High School	Non-hispanic/ latino	803552	33.0	141.1	5.9	50.7
At risk	Spanish	25-34	N/A	Female	High School	Hispanic/ latino	158761	6.5	137.2	5.8	39.2

Clear performance gaps across segments, with **English-speaking males** performing best and **Spanish-speaking females** being the most at risk

Demographic Factors Influence GED Test Outcomes

Cluster Profiles and Key Challenges

Cluster 1 – Higher Performing

- English-speaking
- white
- male
- age 18–24

Potential delay in test-taking despite readiness

Cluster 3 – Lower Performing

- English-speaking
- white
- female
- age 18–24

Lower confidence + higher risk of disengagement

Cluster 2 – Moderate Performing

- English-speaking
- male
- age 15–17

Lack of readiness + unclear preparation path

Cluster 4 – At Risk

- Spanish-speaking
- female
- age 25–34
- Hispanic

Language, access, and preparation gaps

Recommendation 1 - Cluster-Based Personalized Support

Finding: Four distinct learner clusters identified — each requires a tailored pathway to maximize conversion and pass rates.

Cluster 1

High Performing

Use score-based signals to prompt timely test scheduling.
Push automated nudges when GED Ready scores exceed threshold.

Cluster 2

Moderate Performing

Provide guided preparation with diagnostics and clear study pathways.

Cluster 3

Low Performing

Offer personalized practice and targeted feedback to improve weak areas.

Cluster 4

At Risk

Expand Spanish-language support with localized content and bilingual services.

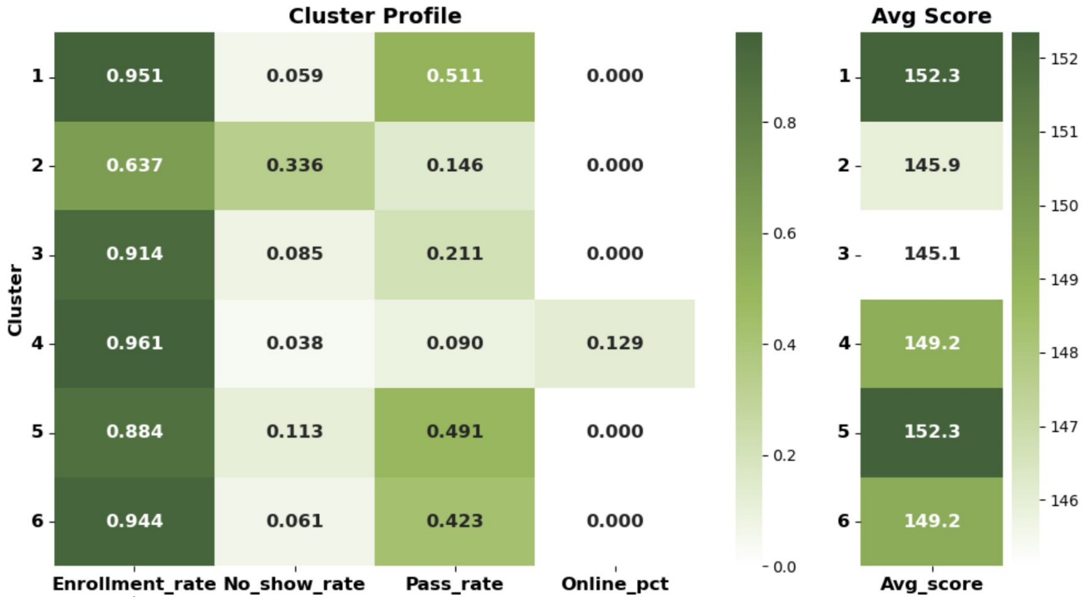
02
Geographic Clustering Analysis

Using K-means Clustering to Identify Low-Performance Geographic Segments

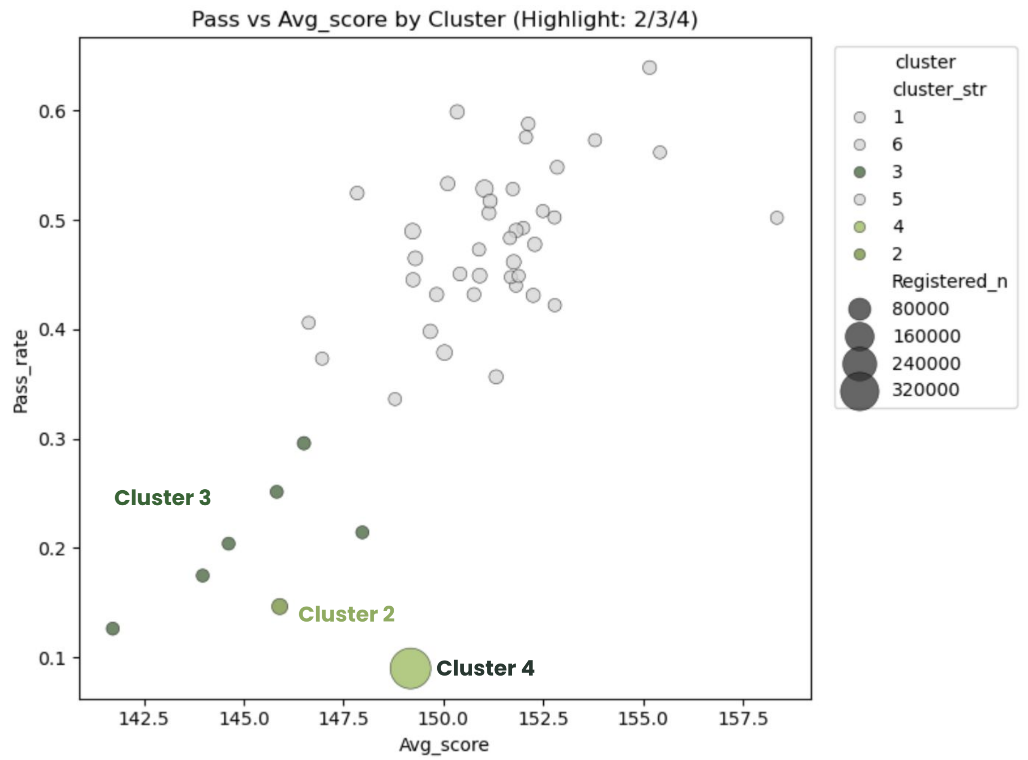
State-Level Indicators

Five standardized metrics capture participation and performance patterns:

- Enrollment Rate
- No-show Rate – percentage of enrolled users who did not attend
- Pass Rate
- Average Score
- Online Percentage – share of online exam delivery



Deep Dive: Cluster 2, Cluster 3 and Cluster 4



Cluster 2

It shows a relatively high no-show rate, and low average scores, which together contribute to its low pass rate.

Cluster 3

It has low pass rates and low average scores, indicating a performance-related issue rather than participation.

Cluster 4

An outlier with low pass rate, likely driven by data concentration or structural bias.

Low Pass Rate Driven by Conversion and Performance Issues

Cluster 2 (NY)

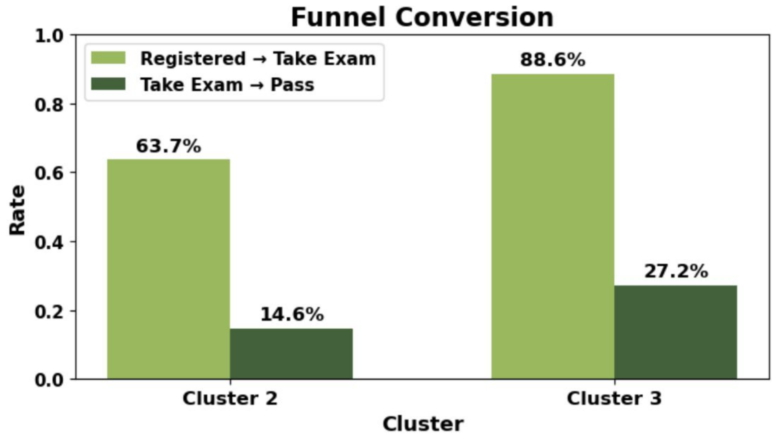
- Large candidate base (21,782 registered candidates)
- It has a much lower registered-to-exam conversion rate(63%).

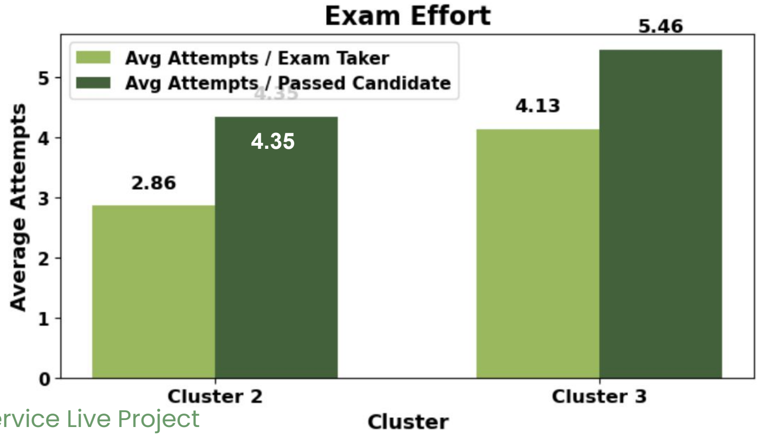
▶ **Implication：** Conversion + Preparation Issue

Cluster 3 (IN, LA, MO, NH, NJ, TN)

- Smaller candidate base (3713 registered candidates)
- It shows stronger participation, but pass performance remains weak.

▶ **Implication：** Mainly performance issue





Cluster 4: Outlier with Large Scale and Extremely Low Pass Rate

Cluster 4 (MN)

Metric	Value
Registered Candidates	365,740
Take Exam Rate	96%
Pass Rate	9%
Avg Attempts / Exam Taker	4.4
Avg Attempts / Passed	9.5

Interpretation

- It behaves fundamentally differently from other clusters
- Strong engagement but significant performance gap
- Indicates potential issues in:
  - Preparation quality
  - Learning support
  - Exam readiness

Recommendation 2 - State-Level Interventions Across Urbanicity

Finding: MN (Urban) shows 5,900+ candidates per center — 10× the national average. Geographic clusters reveal distinct urban vs. rural performance gaps.

Cluster 2

Moderate Conversion

- Improve conversion through automated reminders and booking incentives
- Enhance preparation via online resources or regional prep centers

Cluster 3、Cluster 4

Low Pass Rates

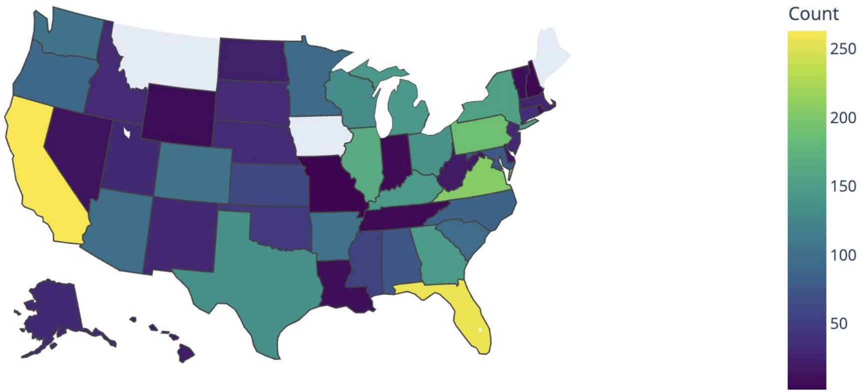
- Increase pass rates through mock exams and personalized learning support
- Target states with persistent multi-subject failures

03 Resource Distribution Analysis

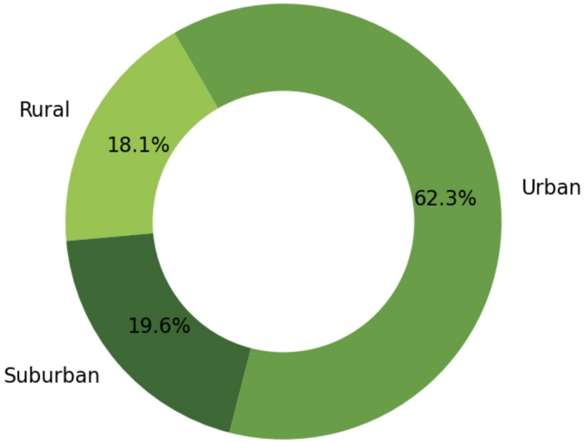
Resource Distribution Landscape

Geographic Distribution of Testing Resources

Distribution of Test Centers Across States



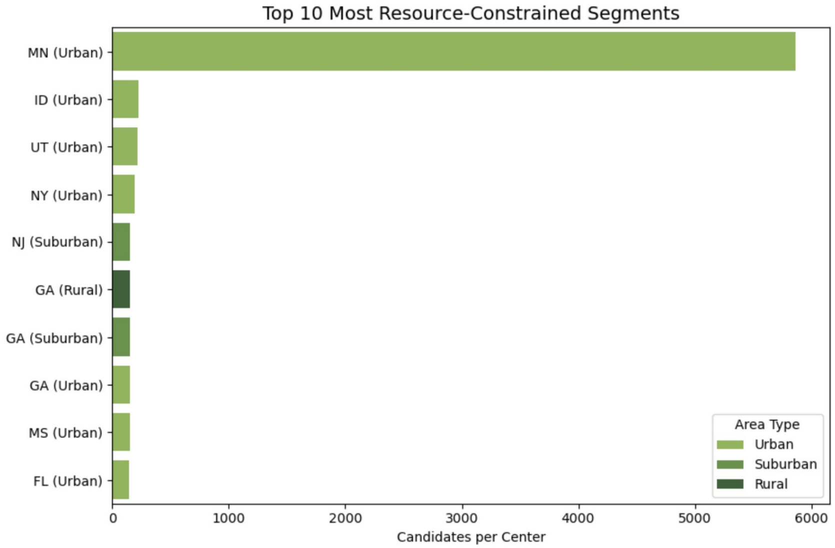
Urban vs Rural Distribution



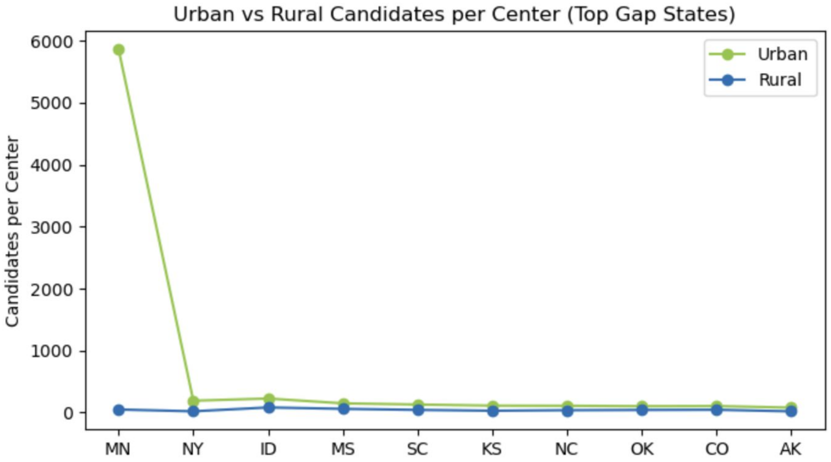
- ❑ Resource allocation follows population density rather than geographic equity
- ❑ Urban concentration(62%) may drive higher congestion in metropolitan areas
- ❑ Rural underrepresentation raises potential accessibility and participation concerns

Resource Pressure & Gap

Candidate per Center Ratio by State



Urban vs Rural Resource Pressure



- **Urban centers consistently face higher candidate loads per center than rural counterparts**, indicating systematic urban resource pressure.
- **Minnesota** is an extreme outlier with disproportionate urban concentration.
- Excluding MN, states such as **Utah (~219), Idaho (~225), and New York (~190)** still exhibit substantial urban congestion.

Driver of Performance Disparities

Infrastructure Stress Factor

To capture structural constraints in testing accessibility, we construct an **Infrastructure Stress Index** using Principal Component Analysis (PCA).

$$InfraFactor = PC_1(Z_{candidate/center}, Z_{urban-rural}, Z_{online})$$

Regression Model

$$PassRate_s = \alpha + \beta \cdot InfraFactor_s + \epsilon_s$$

$$\boxed{\beta = -0.069}$$

Driver Analysis Insight

- **Higher infrastructure pressure (candidate congestion and limited online access) is linked to lower pass rates.**

States such as **Minnesota, Idaho, and Utah** show particularly high candidate-to-center ratios, indicating significant infrastructure pressure.

Recommendation 3 –Rebalancing Infrastructure to Reduce Access Barriers

Finding: Urban centers are severely overloaded while rural areas are underserved. 62% of test centers are urban, yet these serve disproportionately large candidate pools.

#1

Targeted Infrastructure Adjustment

Add testing capacity in top resource-constrained segments. Prioritize MN, ID, UT (Urban) where candidates-per-center far exceeds national benchmarks.

#2

Reallocate Capacity in High-Stress Zones

Partner with community colleges and libraries in saturated urban zones to expand seat availability. Explore mobile testing units for rural areas.

#3

Expand Hybrid Online / Offline Models

Online testing currently accounts for 15% of volume. Increase digital access in underserved regions to reduce geographic dependency on physical centers.

04

Integrated Action Plan

INTEGRATED ACTION SUMMARY

From Insights to Impact

WHO — Demographic

- Target high-performing clusters for scheduling nudges
- Support low performers with personalized Math pathways
- Serve at-risk learners with bilingual & Spanish content

WHERE — Geographic

- Prioritize MN, ID, UT urban overload states
- Cluster 3 & 4 states need mock exams + diagnostics
- Bridge urban-rural gap with targeted state programs

HOW — Resources

- Add physical centers in top 10 constrained segments
- Mobile & community partnerships for rural coverage
- Scale online testing to 30%+ of total volume